

Craig Ssemakula

COMP4358 - Cloud Computing

Professor Azamhet

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This report documents Part II of the Comprehensive Project, where I built a multi-region web application across Canada Central and East US. Each region runs two Linux VMs behind an Application Gateway acting as a regional load balancer, and a single Traffic Manager profile with Geographic routing fronts both regions, sending Canadian visitors to the Canadian Application Gateway and US visitors to the US Application Gateway. All four VMs serve customized Apache pages identifying their region and which VM responded.

1. Architecture Overview

The system has two parallel regional stacks tied together by a Traffic Manager profile (craigproject) that routes by visitor geography. Canada_VNet (10.1.0.0/16) hosts CA_VM1 and CA_VM2 in Canada_VM_Subnet, with Canada_AppGW in its own subnet (Canada_AppGW_Subnet) load balancing between them. US_VNet (10.2.0.0/16) mirrors the same pattern in East US.

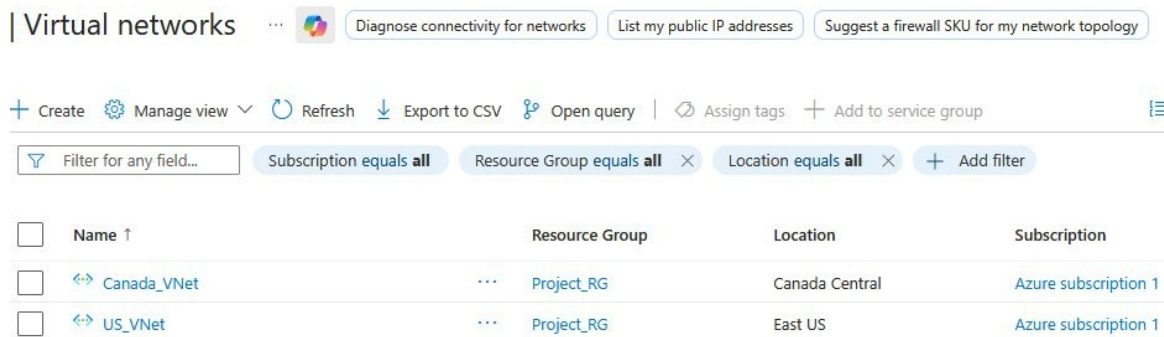
Each region also has its own NAT Gateway (Canada_NATGW and US_NATGW) attached to the VM subnet to provide outbound internet for the VMs at install time. This was set up before deploying the VMs, which avoided the outbound connectivity problem from Part I and let Apache install successfully on the first try for all four VMs.

Traffic Manager sits at the DNS layer. The hostname craigproject.trafficmanager.net resolves differently depending on the requester's geographic location, returning the Canadian Application Gateway's public IP for Canadian users and the US Application Gateway's public IP for US users.

2. Configuration Steps

2.1 Virtual Networks and Subnets

I created two VNets, one per region, each with two subnets: a VM subnet for the Apache servers and a separate Application Gateway subnet (Application Gateways must live in a dedicated subnet within their VNet). Address spaces were 10.1.0.0/16 for Canada and 10.2.0.0/16 for the US to avoid any overlap.



Virtual networks				
Diagnose connectivity for networks List my public IP addresses Suggest a firewall SKU for my network topology				
+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group				
Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter				
☐	Name ↑		Resource Group	Location
☐	↔ Canada_VNet	...	Project_RG	Canada Central
☐	↔ US_VNet	...	Project_RG	East US
				Azure subscription 1

2.2 NAT Gateways for Outbound Connectivity

Before deploying VMs I created Canada_NATGW and US_NATGW, attached a public IP to each (Canada_NATGW_IP and US_NATGW_IP), and associated each NAT Gateway with the corresponding VM subnet. This way the VMs had working outbound internet from the moment they came online.

2.3 Virtual Machines

Four Ubuntu VMs were deployed: CA_VM1 and CA_VM2 in Canada Central, US_VM1 and US_VM2 in East US. None of them have public IPs since all inbound traffic flows through the Application Gateways and outbound traffic flows through the NAT Gateways.

Name ↑		Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP ad
 CA-VM1	...	Azure subscript...	Project_RG	Canada Central	Running	Linux	Standard_L2aos...	-
 CA-VM2	...	Azure subscript...	PROJECT_RG	Canada Central	Running	Linux	Standard_L2aos...	-
 US-VM1	...	Azure subscript...	Project_RG	East US	Running	Linux	Standard_L2aos...	-
 US-VM2	...	Azure subscript...	Project_RG	East US	Running	Linux	Standard_L2aos...	-

2.4 Apache Installation and Custom Pages

Using the Serial Console for each VM, I installed Apache and replaced the default index.html with a region-specific page identifying my name, the course, the region, and which VM is responding:

- CA_VM1: Welcome from Canada - VM1, Region: Canada Central
- CA_VM2: Welcome from Canada - VM2, Region: Canada Central
- US_VM1: Welcome from United States - VM1, Region: East US
- US_VM2: Welcome from United States - VM2, Region: East US

Each page also uses a slightly different background color so it is visually obvious which specific VM the Application Gateway routed the request to.

2.5 Application Gateways

Each region has its own Application Gateway acting as a regional load balancer between its two VMs. Canada_AppGW lives in Canada_AppGW_Subnet and US_AppGW lives in US_AppGW_Subnet. Both are Standard V2 with a single instance and a basic HTTP listener on port 80.

Target type	Target
Virtual machine	ca-vm1905_z1
Virtual machine	ca-vm2917_z1

The Canadian backend pool (Canada_BackendPool) targets CA_VM1 and CA_VM2 directly:

Name ↑		Type	Resource Group	Location
 Canada_AppGW	...	Application gateway	Project_RG	Canada Central
 US_AppGW	...	Application gateway	Project_RG	East US

The US backend pool (US_BackendPool) targets US_VM1 and US_VM2 directly:

Target type	Target
Virtual machine	us-vm1586_z1
Virtual machine	us-vm2133_z1

Each Application Gateway also has a basic routing rule (Canada_Rule and US_Rule) that ties the public listener to its regional backend pool through HTTP backend settings on port 80.

2.6 DNS Name Labels

Both Application Gateway public IPs were given DNS name labels (craigproject-ca and craigproject-us). Traffic Manager will not list a public IP as a selectable endpoint unless the IP has a DNS label, so this step was required before the next phase.

2.7 Traffic Manager Profile with Geographic Routing

I created a Traffic Manager profile named craigproject with the routing method set to Geographic. The profile is reachable as craigproject.trafficmanager.net.

| Configuration ☆ ...

ems

Save Discard

Routing method ⓘ

Geographic

DNS time to live (TTL)* ⓘ

60 seconds

Endpoint monitor settings ⓘ

Protocol

HTTP

Port *

80

Two Azure endpoints were added to the profile, each pointing to one of the regional Application Gateway public IPs and mapped to its target country:

- Canada_Endpoint -> Canada_AppGW_IP, geo-mapped to Canada
- US_Endpoint -> US_AppGW_IP, geo-mapped to United States

Both endpoints became Online once Traffic Manager finished its first round of HTTP probes against the Application Gateway public IPs:

craigproject | Endpoints ☆ ...

Traffic Manager profile

search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Links

+ Add Refresh Delete

Search endpoints

Name ⓘ	Status ⓘ	Monitor Status ⓘ	Type ⓘ
Canada_Endpoint	Enabled	Online	Azure endpoint
US_Endpoint	Enabled	Online	Azure endpoint

3. Verification

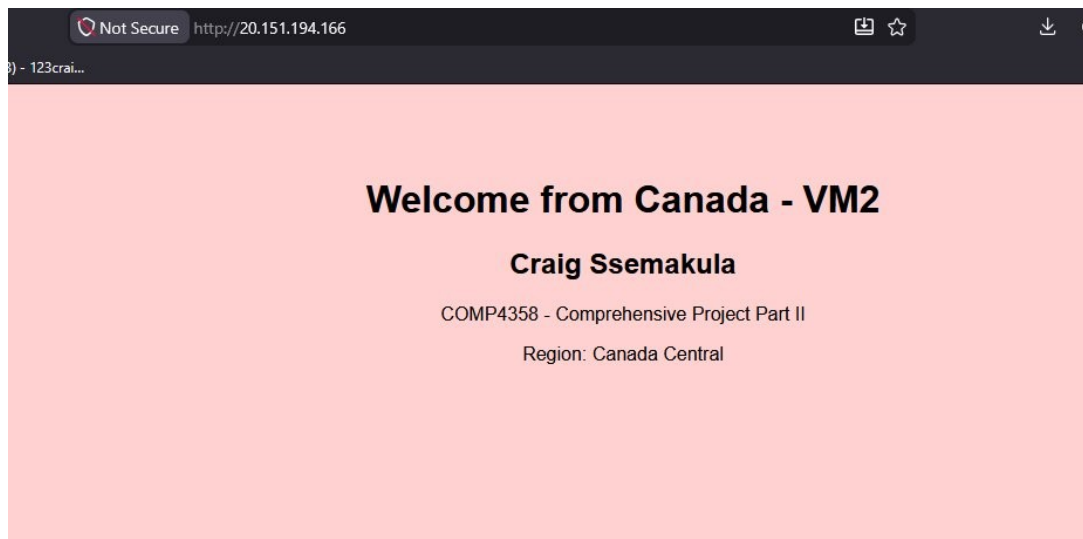
Verification was done in three steps: confirm each Application Gateway routes correctly within its own region, then confirm Traffic Manager routes correctly by geography for both a US source IP and a Canadian source IP.

3.1 Application Gateways Working Directly

Browsing to the US Application Gateway's public IP (52.188.37.141) returned a US VM page, proving the US Application Gateway is reachable and routing to its backend pool:



Browsing to the Canadian Application Gateway's public IP (20.151.194.166) returned a Canadian VM page, confirming the Canadian side works the same way:



3.2 Traffic Manager from a US IP

From my real network connection (Comcast Cable Communications, Greenwood Village, Colorado), the IP geolocation correctly identifies the country as United States:

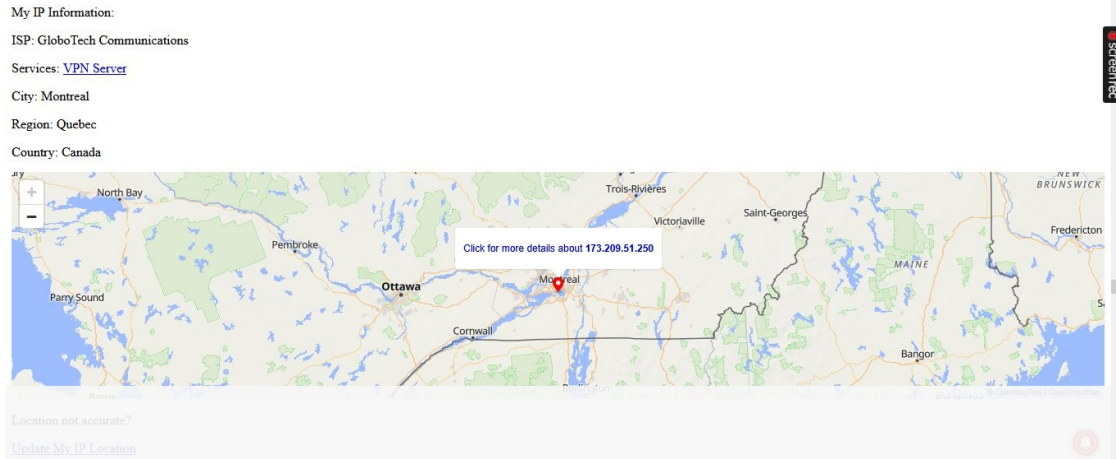


Browsing to craigproject.trafficmanager.net from this US IP returned a US VM page, confirming Traffic Manager directed the request to US_AppGW based on the source country:

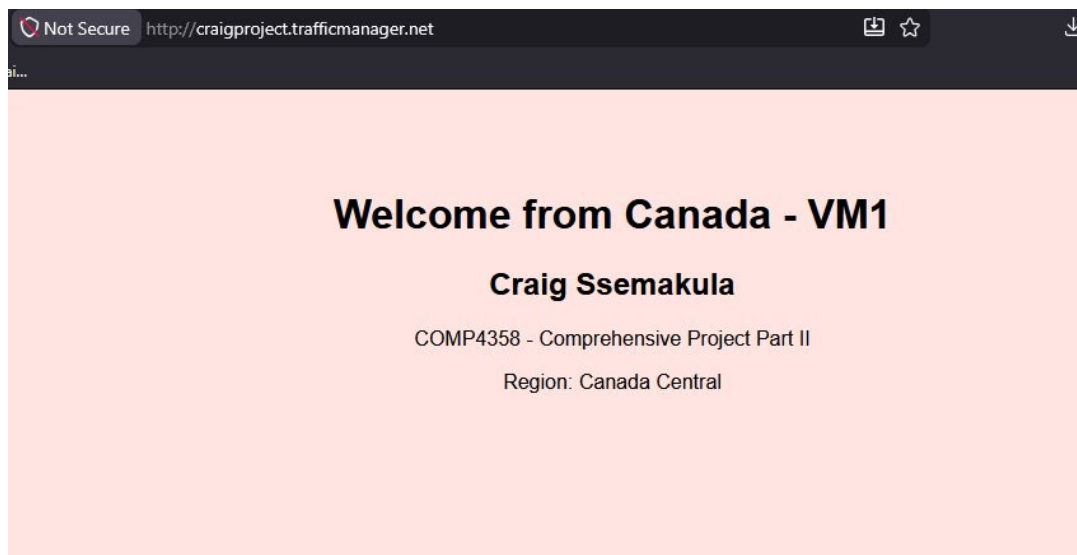


3.3 Traffic Manager from a Canadian IP

Using a VPN connection terminating in Montreal, Quebec, my apparent source IP became Canadian (173.209.51.250):

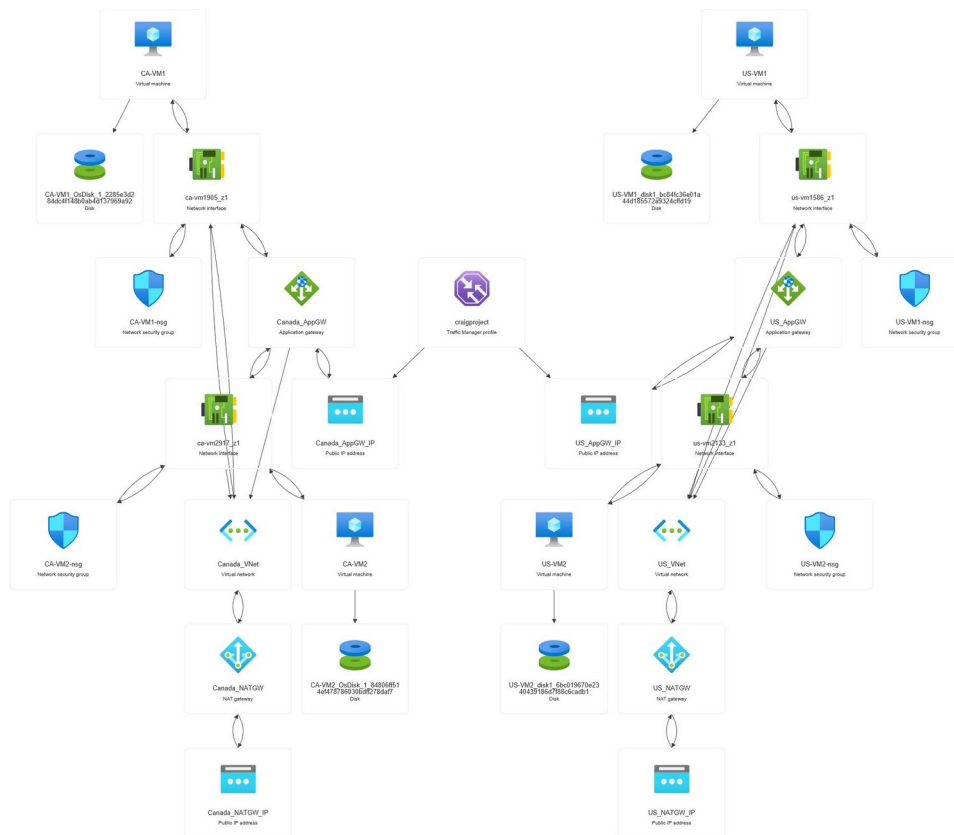


Reloading `craigproject.trafficmanager.net` in a fresh browser window from the Canadian IP returned a Canadian VM page, confirming geographic routing redirected the same domain to Canada_AppGW:



4. Resource Topology

The Resource Visualizer for Project_RG shows the full multi-region setup: each region with its VNet, two VMs, two NSGs, two NICs, an Application Gateway with its own public IP, and a NAT Gateway with its own public IP. The Traffic Manager profile (`craigproject`) sits in the middle and connects to both Application Gateway public IPs.



5. Challenges and Solutions

5.1 Avoiding the Outbound Internet Issue from Part I

In Part I I ran into a problem where the VMs could not reach the apt repositories because the Standard Load Balancer did not provide default outbound connectivity, which only got fixed after attaching a NAT Gateway. For Part II I created Canada_NATGW and US_NATGW with their public IPs attached and associated to the VM subnets before any VM was deployed. As a result, apt update and apt install apache2 worked on the first attempt for all four VMs.

5.2 DNS Name Label Required for Traffic Manager Endpoint Selection

When I first opened the endpoint creation panel for the Traffic Manager profile, the dropdown for Public IP did not show either Application Gateway public IP. The fix was to open each public IP, set a DNS name label (craigproject-ca and craigproject-us), and save. After that both IPs appeared in the dropdown and I could finish adding the endpoints.

5.3 Browser DNS Cache During Geographic Testing

After connecting the VPN to Canada, my first attempts at loading craigproject.trafficmanager.net still returned the US page because the browser was reusing a cached DNS answer. Using a fresh incognito window forced a new DNS lookup that picked up the Canadian endpoint.

6. What I Learned

Geographic routing in Traffic Manager works at the DNS level, not at the HTTP level. The user's resolver decides which IP to hand back based on its public source address, and the user's browser then connects directly to that regional Application Gateway. Because of this, the

regional Application Gateways and their VMs are not dependent on each other at runtime, which makes the design more resilient.

Application Gateway sits one layer below Traffic Manager and handles the in-region load balancing. It needs its own subnet, a backend pool, an HTTP listener, backend HTTP settings, and a routing rule that ties the listener to the backend pool. Once those exist, hitting the gateway's public IP cycles requests through the backend VMs based on its health probes.

Setting up outbound connectivity (NAT Gateway with attached public IP) before deploying the VMs is much smoother than fixing it afterwards. Doing it up front also avoids partial failures where some packages install and others fail because the connection drops mid-update.

7. Conclusion

All Part II requirements were met. Two regional stacks (Canada Central and East US) each run two Apache VMs behind an Application Gateway, and a single Traffic Manager profile (craigproject.trafficmanager.net) directs visitors to the correct regional gateway based on their source country. Verification from a US IP returned a US VM page, and verification from a Canadian VPN IP returned a Canadian VM page, both through the same domain.